

Maximilian Gorsky

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Age	born in 1993	Mail Address	Discrete Mathematics Group
Languages	English, German		Institute for Basic Science (IBS)
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Nationality	German		Republic of Korea

Research Interests

- **Algorithmic Structural Graph Theory.** Especially Graph Minor Structure Theory as established by Robertson and Seymour, but also generalisations of these concepts to directed graphs. All with a view towards using graph structure to solve hard algorithmic problems.
- **Matching Theory.** With a focus on the Structural Matching Theory of bipartite graphs and applications to the forcing number, as well as various Hamiltonicity Problems.
- **Parameterized Algorithms.** Including relations between model checking and structural graph theory.

Work History

Nov 2024 - Present	Institute for Basic Science, Daejeon, South Korea <i>Senior Researcher</i> Research in the areas of (di)graph structure theory, matching theory, and algorithmics in the DIMAG group headed by CI Sang-il Oum.
Mar 2026 - Jun 2026	KAIST, Daejeon, South Korea <i>Adjunct Professor</i> Teaching a course on algorithmic graph theory (CS49900E) in the spring semester at the School of Computing in KAIST to a group of roughly 30 students.
Sep 2019 - Oct 2024	Technical University of Berlin, Germany <i>Research & Teaching Assistant, Course Coordinator</i> Research in the areas of (di)graph structure theory, matching theory, and algorithmics. Teaching small and large groups (up to 450 students), creating learning material, providing consultation hours, managing student tutors (including hiring), and organising exams in modules for logic, discrete structures, algorithmics, (structural) graph theory, and complexity theory for Bachelor and Master students, as well as supervising Bachelor's and Master's theses.
Oct 2016 - Sep 2019	Technical University of Berlin, Germany <i>Student Tutor</i> Teaching small groups, providing consultation hours, and aiding in the organisation of exams.

Education

Sep 2019- Sep 2024	Ph.D. - Technical University of Berlin (graduated summa cum laude) at the Institute of Software Engineering and Theoretical Computer Science Thesis Title: The Structure of (Even) Directed Cycles Advisor: Stephan Kreutzer
Oct 2017- Sep 2019	M.Sc. in Computer Science - Technical University of Berlin Master's Thesis: Towards a Characterisation of König Graphs
Oct 2012- Mar 2018	B.Sc. in Computer Science - Technical University of Berlin Bachelor's Thesis: Bricks of Small Connectivity

Publications

Conference publications

1. Dario Cavallaro, Maximilian Gorsky, Stephan Kreutzer, Dimitrios Thilikos, and Sebastian Wiederrecht. Optimal Bounds for the k -Disjoint Paths Problem, May 2026. doi:10.48550/arXiv.2605.14902 [accepted to **FOCS** 2026]
2. Maximilian Gorsky, Evangelos Protopapas, and Sebastian Wiederrecht. Quickly excluding an annotated planar graph, February 2026. arXiv:2602.06516 [accepted to **ICALP** 2026]
3. Maximilian Gorsky, Michał T. Seweryn, and Sebastian Wiederrecht. The price of homogeneity is polynomial, February 2026. arXiv:2602.01882 [accepted to **ICALP** 2026]
4. Mujin Choi, Maximilian Gorsky, Gunwoo Kim, Caleb McFarland, and Sebastian Wiederrecht. Odd-Cycle-Packing-treewidth: On the Maximum Independent Set problem in odd-minor-free graph classes, November 2025. arXiv:2511.10019 [accepted to **ICALP** 2026]
5. Maximilian Gorsky, Giannos Stamoulis, Dimitrios M. Thilikos, and Sebastian Wiederrecht. Catching Rats in H -minor-free Graphs. In *Proceedings of the 2026 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 4644–4659, Vancouver, Canada, January 2026. Society for Industrial and Applied Mathematics. doi:10.1137/1.9781611978971
6. Maximilian Gorsky, Michał T. Seweryn, and Sebastian Wiederrecht. Polynomial bounds for the Graph Minor Structure Theorem. In *2025 IEEE 66th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 1961–1978, Sydney, Australia, December 2025. IEEE. doi:10.1109/FOCS63196.2025.00104
7. Maximilian Gorsky, Ken-ichi Kawarabayashi, Stephan Kreutzer, and Sebastian Wiederrecht. Packing Even Directed Circuits Quarter-Integrally. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC)*, pages 692–703, Vancouver BC Canada, June 2024. ACM. doi:10.1145/3618260.3649682
8. Jakub Gajarský, Maximilian Gorsky, and Stephan Kreutzer. Differential Games, Locality, and Model Checking for FO Logic of Graphs. In *30th EACSL Annual Conference on Computer Science Logic (CSL 2022)*, volume 216, pages 22:1–22:18. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2022. doi:10.4230/LIPIcs.CSL.2022.22
9. Maximilian Gorsky, Raphael Steiner, and Sebastian Wiederrecht. Strongly Pfaffian Graphs. In *Extended Abstracts EuroComb 2021*, volume 14, pages 267–272. Springer International Publishing, Cham, 2021. doi:10.1007/978-3-030-83823-2_42

Journal publications

1. Katie Clinch, Sean Dewar, Niloufar Fuladi, Maximilian Gorsky, Tony Huynh, Eleftherios Kastas, Atsuhiko Nakamoto, Anthony Nixon, and Brigitte Servatius. Triangulated Spheres with Holes in Triangulated Surfaces. *Discrete & Computational Geometry*, March 2026. doi:10.1007/s00454-026-00836-8
2. Maximilian Gorsky and Fabian Kreßin. Computing the forcing spectrum of outerplanar graphs in polynomial time. *Discrete Applied Mathematics*, 377:204–217, December 2025. doi:10.1016/j.dam.2025.06.063
3. Maximilian Gorsky, Theresa Johanni, and Sebastian Wiederrecht. A note on the 2-factor Hamiltonicity Conjecture. *Discrete Mathematics*, 348(6):114442, June 2025. doi:10.1016/j.disc.2025.114442
4. Maximilian Gorsky, Raphael Steiner, and Sebastian Wiederrecht. Matching theory and Barnette's conjecture. *Discrete Mathematics*, 346(2):113249, February 2023. doi:10.1016/j.disc.2022.113249

Manuscripts & Preprints

1. J. Pascal Gollin, Maximilian Gorsky, Meike Hatzel, Kevin Hendrey, Tony Huynh, Caleb McFarland, Marek Sokołowski, Sebastian Wiederrecht, and Paul Wollan. An Erdős-Pósa theorem for cycles and faces of distinct lengths, July 2026. doi:10.48550/arXiv.2607.06869
2. Maximilian Gorsky, Kevin Hendrey, and Tony Huynh. The Erdős-Pósa property for prime-length cycles fails (and beyond), May 2026. arXiv:2605.04938, doi:10.48550/arXiv.2605.04938 [submitted]

3. Maximilian Gorsky and Clemens Kuske. On non-planar, cycle-conformal graphs, February 2026. arXiv: 2602.07331 [submitted]
4. Maximilian Gorsky and Michał T. Seweryn. Posets with k -outerplanar cover graphs have bounded dimension, May 2021. arXiv:2103.15920

PhD Thesis

1. Maximilian Gorsky. *The Structure of (Even) Directed Cycles*. PhD thesis, Technische Universität Berlin, September 2024. doi:10.14279/depositonce-21276

Awards

- 2025 IBS Researcher of the Year Award

Research Visits

- **10.–28. August 2026** TU Berlin (Host: Stephan Kreutzer)
- **27. June–04. July 2026** GeorgiaTech (Host: Rose McCarty)
- **17.–18. November 2025** TU Vienna (Host: Jan Dreier)
- **24. August–06. September 2025** McGill University (Host: Sergey Norin)
- **17. July–05. August 2025** TU Berlin (Host: Stephan Kreutzer)
- **22. April–31. May 2024** IBS-DIMAG (Host: Sang-il Oum)
- **04.–10. March 2022** LIRMM (Host: Dimitrios Thilikos)

Selected Talks

- 28.09.2026 (**GROW**, Będlewo, Poland) *Controlling major apices in the polynomial Graph Minor Structure Theorem*
- 24.06.2026 (**Invited**) (**SIAM Conference on Discrete Mathematics (DM26)**, San Diego, USA) *Controlling major apices in the polynomial Graph Minor Structure Theorem*
- 06.05.2026 (Discrete Math Seminar, Daejeon, South Korea) *The Disjoint Paths Problem lies in the Oort cloud of algorithms*
- 13.01.2026 (**SODA**, Vancouver, Canada) *Catching Rats in H -minor-free Graphs*
- 19.11.2025 (**LoGAlg**, Vienna, Austria) *Catching Rats in H -minor-free Graphs*
- 20.09.2025 (**Invited**) (**KPPY100**, Gyeongju, South Korea) *The Grid Theorem in H -minor-free graphs*
- 19.08.2025 (**Invited**) (**Koper-Leipzig Graph Theory Workshop**, Koper, Slovenia) *Where to go next in Graph Minor Theory?*
- 23.07.2025 (Mittagsseminar, Berlin, Germany) *On a variant of the crossing number in the graph minor structure theorem*
- 22.05.2025 (**CanaDAM**, Ottawa, Canada) *Polynomial bounds for the Graph Minor Structure Theorem*
- 19.12.2024 (**Invited**) (**32nd KIAS Combinatorics Workshop**, Busan, South Korea) *Polynomial bounds for the Graph Minor Structure Theorem*
- 11.09.2024 (**GROW**, Cottbus, Germany) *Polynomial bounds for the Graph Minor Structure Theorem*
- 25.06.2024 (**STOC**, Vancouver, Canada) *Packing even directed circuits quarter-integrally*
- 30.04.2024 (Discrete Math Seminar, Daejeon, South Korea) *Towards packing even dicycles half-integrally*
- 05.07.2022 (**ICGT**, Montpellier, France) *Matching Theory and Barnette's Conjecture*
- 25.02.2022 (Mittagsseminar, Berlin, Germany) *Matching Theory and Barnette's Conjecture*
- 16.02.2022 (**CSL**, Online) *Differential games, locality, and model checking for FO logic of graphs*
- 09.09.2021 (**EUROCOMB**, Online) *Strongly Pfaffian Graphs*
- 07.05.2021 (Mittagsseminar, Online) *k -Outerplanarity and Poset Dimension*

Invited to Workshops

- Second 2026 **Barbados Graph Theory Workshop**, 18.-24.12.2026, Holetown, Barbados
- **GROW** 27.09.-01.10.2026, Będlewo, Poland
- **MATRIX: Global Structure and Geometry of Graphs** 13.-24.04.2026, Melbourne, Australia
- **5th East Asia Workshop on Extremal and Structural Graph Theory** 27.-30.11.2025, Seoul, South Korea
- **LoGAlg** 19.-21.11.2025, Vienna, Austria
- **Bertinoro Workshop on Algorithms and Graphs 2025** 26.-31.10.2025, Bertinoro, Italy
- **KPPY100** 19.-21.09.2025, Gyeongju, South Korea
- **Koper-Leipzig Graph Theory Workshop** 18.-22.08.2025, Koper, Slovenia
- **32nd KIAS Combinatorics Workshop** 19.-21.12.2024, Busan, South Korea
- **GROW** 09.-12.09.2024, Cottbus, Germany
- **SGCW 2023 on Chi-Boundedness** 27-31.03.2023, Online
- **GROW** 19.-22.09.2022, Koper, Slovenia
- **Sparse Graph Coalition Workshop (SGCW) 2022 on Digraphs** 04.-08.04.2022, Online
- **Order & Geometry** 09.-13.09.2020, Wittenberg, Germany
- **GRASTA** 24.-27.09.2018, Berlin, Germany

Journal and Conference Refereeing

Journals Discrete Mathematics($\times 2$), Discussiones Mathematicae Graph Theory,
Journal of Combinatorial Theory Series B($\times 2$), Electronic Journal of Combinatorics($\times 2$)

Conferences FOCS($\times 3$), SODA($\times 3$), WG($\times 3$), MFCS($\times 2$), IPEC, IWOCa, LICS, WADS

Additional Teaching

■ Winter School on Graph Minors in February 2025

Organisation and teaching in the context of the Winter School on Graph Minors at the IBS Headquarters in Daejeon, South Korea from 03.02. to 19.02.2025. Taught 9 of 18 lectures, each lasting at least 90 minutes, to a group of roughly 35 students, graduate students, and senior academics. The topics of the lectures were the most current techniques and results in the theory of Graph Minors available at the time.

Social Engagement

■ Long Night of Sciences (LNDW) in 2017, 2018, and 2019

Participation in the organisation and running of an escape room, a rally, and a public display themed around theoretical computer science for the Long Night of Sciences (LNDW) in Berlin.

■ Membership in professional organisations

Associate Fellow of the ICA since Nov 2025, member of the KMS since Nov 2024

Supervised Theses

- **Fanny Hauser** Cops, Robbers, and Cycle Degree (*Bachelor*) 2022
- **Fabian Kreßin** Forcing Numbers on Outerplanar Graphs (*Bachelor*) 2023
- **Clemens Kuske** Towards a Characterization of Cycle Conformal Graphs (*Bachelor*) 2024
- **Theresa Johanni** 2-Factor Hamiltonicity and Matching Theory (*Bachelor*) 2024
- **Julius Wachlin** Explicitly Computing the Tight Cut Decomposition (*Bachelor*) 2024
- **Fabian Kreßin** Exploring the Parameterised Complexity of Forcing Numbers (*Master*) 2025

Date compiled: October 6, 2026